

Automating Everything with Ansible

This practical walkthrough of how Ansible works gives an insight into why it excels at configuration management and IT automation.



Ansible, created by Michael DeHaan, is an open source IT automation engine that simplifies the orchestration of tasks such as software deployment and system configuration. Other advanced tasks like continuous deployment and rolling updates with zero-downtime are also supported by this automation engine. Ansible was released in 2012 and is currently a Red Hat product and part of the Ansible Automation Platform.

Ansible does not need in-built agents and custom security infrastructures like other automation tools. It uses SSH (or WinRM for Windows) to communicate with the target system. Then, Ansible copies modules, executes them, and removes them. Nothing is left on the target system. As a result of this architecture, Ansible has a low maintenance cost when it is used throughout an enterprise.

Ansible architecture

Ansible's architecture favours simplicity and ease of maintenance. For example, all the models that it manages are passive. As a result of this design choice, the control node does all the thinking and contains everything. Inventories (lists of hosts), playbooks (YAML files), modules (files that support tasks), and plugins (files that support enhancements) are all located on the control node. The *ansible* and *ansible-playbook* commands provide control of all these files and

resources. These commands also support a Python API.

Managed nodes are the servers being configured. There is no agent, daemon, or persistent process to install on them — only an SSH server and a Python interpreter. When a playbook runs, Ansible connects over SSH (or WinRM for Windows), copies the relevant module code to the host as a small script, executes it, captures the result, and removes it. Nothing is left running afterward, which is what makes the model 'agentless' and lets a new managed node be brought under Ansible's control in minutes.

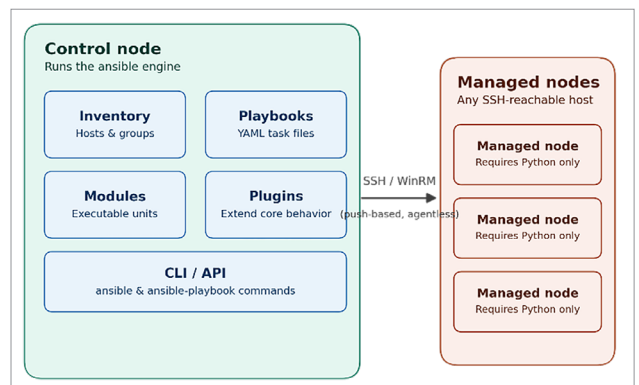


Figure 1: Control node components pushing configuration to manage nodes over SSH